

Computational perspectives on the Classical Decision Problem

Lecture 9: The Fluted Fragment

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$$\forall x_1 (\text{lectr}(x_1) \rightarrow \neg \exists x_2 (\text{prof}(x_2) \wedge \forall x_3 (\text{stdt}(x_3) \rightarrow \text{intro}(x_1, x_2, x_3))))).$$

- Think of arguments of predicates as being 'lined up' in columns:

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The allusion here is architectural:



- The following formulas are not in the fluted fragment:

$$\forall x_1 r(x_1, x_1)$$

$$\forall x_1 \forall x_2 (r(x_1, x_2) \rightarrow r(x_2, x_1))$$

$$\forall x_1 \forall x_2 (r(x_1, x_2) \rightarrow \forall x_3 (r(x_2, x_3) \rightarrow r(x_1, x_3)))$$

- On the other hand if there is at most one variable, fluting makes no difference.

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- For all $m \geq 0$, a **fluted m -atom** is an expression $p(x_\ell, \dots, x_m)$, where $\ell \leq m+1$, p is a predicate of arity $m-\ell+1$ (not equality), and $x_\ell, \dots, x_m$ is a contiguous sub-sequence of $\bar{x}_\omega$.

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- We define the sets of formulas $\mathcal{FL}^{[m]}$ (for $m \geq 0$) by simultaneous structural induction as follows:
 - any fluted m -atom is a formula of $\mathcal{FL}^{[m]}$;
 - $\mathcal{FL}^{[m]}$ is closed under Boolean combinations;
 - if φ is in $\mathcal{FL}^{[m+1]}$, then $\exists x_{m+1} \varphi$ and $\forall x_{m+1} \varphi$ are in $\mathcal{FL}^{[m]}$.

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 - if φ is in $\mathcal{FL}^{[m+1]}$, then $\exists x_{m+1} \varphi$ and $\forall x_{m+1} \varphi$ are in $\mathcal{FL}^{[m]}$.
- The set of *fluted formulas* is defined as $\mathcal{FL} = \bigcup_{m \geq 0} \mathcal{FL}^{[m]}$.

- Denote by $\mathcal{FL}^m$ the subfragment of $\mathcal{FL}$ consisting of those formulas featuring at most m variables, free or bound, that is:

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$$\forall x_1 (\text{lectr}(x_1) \rightarrow \neg \exists x_2 (\text{prof}(x_2) \wedge \forall x_3 (\text{stdt}(x_3) \rightarrow \text{intro}(x_1, x_2, x_3))))$$

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is in $\mathcal{FL}^m$ for all $m \geq 3$, but is in $\mathcal{FL}^{[m]}$ only for $m = 0$.

- For quantifier-free formulas, there is no difference between $\mathcal{FL}^m$ and $\mathcal{FL}^{[m]}$.

- An $\mathcal{FL}^m$ -formula is in (fluted) normal-form if it has the shape

$$\bigwedge_{i \in I} \forall x_1 \cdots \forall x_{m-1} (\mu_i \rightarrow \exists x_m \gamma_i) \wedge \bigwedge_{j \in J} \forall x_1 \cdots \forall x_{m-1} (\nu_j \rightarrow \forall x_m \delta_j),$$

where $m \geq 1$, I and J are finite sets of indices, the μ_i and ν_j are quantifier-free $\mathcal{FL}^{m-1}$ -formulas, and the γ_i and δ_j are quantifier-free $\mathcal{FL}^m$ -formulas.

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Lemma

Let φ be an $\mathcal{FL}^m$ -sentence, $m \geq 1$. We can compute, in time bounded by a polynomial function of $\|\varphi\|$, a normal-form $\mathcal{FL}^m$ -formula satisfiable over the same domains as φ .

The proof proceeds via re-writing techniques similar to those used for the normal-form lemmas for $\mathcal{FO}^2$ and $\mathcal{GF}$.

- Fluting allows us to delete variables from formulas entirely:

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$$\forall(\text{lectr} \rightarrow \neg \exists(\text{prof} \wedge \forall(\text{stdt} \rightarrow \text{intro}))).$$

- This includes, normal-form formulas, thus:

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- which is more compact.

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- We shall show that, for any $\mathcal{FL}^m$ -sentence φ ($m \geq 3$), we can compute an $\mathcal{FL}^{m-1}$ -sentence ψ , such that: (i) φ is satisfiable if and only if ψ is satisfiable; (ii) φ is finitely satisfiable if and only if ψ is finitely satisfiable.

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- (Actually, we can improve on this very slightly.)

- To see how the (exponential) reduction from $\text{Sat}(\mathcal{FL}^m)$ to $\text{Sat}(\mathcal{FL}^{m-1})$, works, let us take a sentence φ of $\mathcal{FL}^m$, which we may assume to be in normal form:

$$\bigwedge_{i \in I} \forall^{m-1}(\mu_i \rightarrow \exists \gamma_i) \wedge \bigwedge_{j \in J} \forall^{m-1}(\nu_j \rightarrow \forall \delta_j)$$

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- For all $i \in I$ and all $J' \subseteq J$, let $p_{i,J'}$ and $q_{J'}$ be new predicates of arity $m-2$.
- Intended interpretations:
 - $p_{i,J'}$ characterizes those $(m-2)$ -tuples $\bar{a}$ such that, for some a , the $(m-1)$ -tuple $a, \bar{a}$ satisfies $\mu_i \wedge \bigwedge_{j \in J'} \nu_j$.
 - $q_{J'}$ characterizes those $(m-2)$ -tuples $\bar{a}$ such that, for some object a , the $(m-1)$ -tuple $a, \bar{a}$ satisfies $\bigwedge_{j \in J'} \nu_j$.

Lemma

Let χ be a quantifier-free $\mathcal{FL}^m$ -formula in some signature Σ and let $\Sigma^- \subseteq \Sigma$. Denote by χ° the strongest quantifier-free $\mathcal{FL}^m$ -formula in the signature Σ^- entailed by χ . Then χ° can be computed in time bounded by an exponential function of $\|\chi\|$. Moreover, if τ^- is a fluted m -type over Σ^- such that $\models \tau^- \rightarrow \chi^\circ$, then τ^- can be extended to a fluted m -type $\tau \supseteq \tau^-$ over Σ such that $\models \tau \rightarrow \chi$.

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- In the sequel, we will be interested in the case where Σ^- consists of all the predicates of Σ of arity **strictly** less than m .
- Thus, if χ is a quantifier-free formula of $\mathcal{FL}^m$, then χ° cannot involve any literals featuring m -ary predicates.

- Remembering that

$$\varphi := \bigwedge_{i \in I} \forall^{m-1}(\mu_i \rightarrow \exists \gamma_i) \wedge \bigwedge_{j \in J} \forall^{m-1}(\nu_j \rightarrow \forall \delta_j),$$

our formula ψ will comprise 4 groups of conjuncts.

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- Conjuncts fixing the interpretations of the $p_{i,J'}$:

$$\bigwedge_{i \in I} \bigwedge_{J' \subseteq J} \forall^{m-1}((\mu_i \wedge \bigwedge_{j \in J'} \nu_j) \rightarrow p_{i,J'})$$

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- Conjuncts securing the witness requirements of φ :

$$\bigwedge_{i \in I} \bigwedge_{J' \subseteq J} \forall^{m-2}(p_{i,J'} \rightarrow \exists(\gamma_i \wedge \bigwedge_{j \in J'} \delta_j)^\circ)$$

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- However, φ_2 is exponentially larger than φ_1 , over an exponentially larger signature.
- Iterating, we eventually obtain a formula φ_{m-1} in $\mathcal{FL}^2$ such that φ_{m-1} is (finitely) satisfiable if and only if φ_1 is, with φ_2 $(m-2)$ -tuply exponentially larger than φ_1 , over an $(m-2)$ -tuply exponentially larger signature.

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- In that case, $\varphi_3, \dots, \varphi_m$ all have finite models. (These are a little larger, but the increase is small and does not affect orders of exponentials.)
- Thus, if φ is satisfiable, then it has a model of $(m-1)$ -tuply exponential size. We have proved:

Theorem

If $m \geq 2$, then any satisfiable formula of $\mathcal{FL}^m$ has a model of size bounded by an $(m-1)$ -tuply exponential function of $\|\varphi\|$. Hence, $\text{Sat}(\mathcal{FL}^m)$ is in $(m-1)$ -NEXPTIME.

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- It is possible to show, using a construction essentially similar to that employed for $\text{Sat}(\mathcal{FO}^2)$, that any satisfiable formula φ of $\mathcal{FL}^3$ has a model of size bounded by an exponential function of $\|\varphi\|$.

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- It is possible to show, using a construction essentially similar to that employed for $\text{Sat}(\mathcal{FO}^2)$, that any satisfiable formula φ of $\mathcal{FL}^3$ has a model of size bounded by an exponential function of $\|\varphi\|$.
- This allows us to terminate the reduction sequence one step earlier, and we obtain:

Theorem

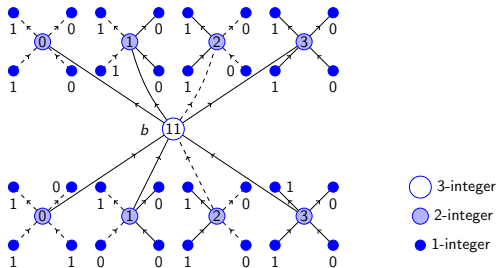
If $m \geq 3$, then any satisfiable formula of $\mathcal{FL}^m$ has a model of size bounded by an $(m-2)$ -tuply exponential function of $\|\varphi\|$. Hence, $\text{Sat}(\mathcal{FL}^m)$ is in $(m-2)$ -NEXPTIME.

- We have an (almost) matching lower bound:

Theorem (Szwast, Tendera, P-H, 2019)

$\text{Sat}(\mathcal{FL}^m)$ is $\lfloor m/2 \rfloor$ -NEXPTIME-hard, for $m \geq 2$.

- Proof strategy: code numbers up to $2^m - 1$ with 'double trres' of height $2m$:



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- Thus, for $m \geq 3$ the complexity of $\text{Sat}(\mathcal{FL}^m)$ lies somewhere between $\lfloor m/2 \rfloor\text{-NEXPTIME}$ and $(m-2)\text{-NEXPTIME}$.

- Thus, $\text{Sat}(\mathcal{FL})$ is non-elementary.
- Thus, for $m \geq 3$ the complexity of $\text{Sat}(\mathcal{FL}^m)$ lies somewhere between $\lfloor m/2 \rfloor$ -NEXPTIME and $(m-2)$ -NEXPTIME.
- At the time of writing, it is not known how to close the complexity gap for $m \geq 5$.

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- We can add counting quantifiers to the fluted fragment and still retain decidability.